Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Willem Postma Dairy #1

Physical address of dairy:

Number and Street	City	County	Zip Code
3700 Claus RD	Modesto	Stanislaus	95357

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1962

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X014-X013-X003-XXXX

B. OPERATORS

Postma, Willem

Operator name: Postma, Willem

Telephone no.: (209) 551-2930 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
3700 Claus RD	Modesto	CA	95357

This operator is responsible for paying permit fees.

C. OWNERS

Postma, Willem

Legal owner name: Postma, Willem

Telephone no.: (209) 551-2930 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
3700 Claus RD	Modesto	CA	95357

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	600	550	200	190
Number under roof	990	160	0	0	0	0
Maximum number	990	160	600	550	200	190
Average number	990	160	600	550	200	190
Avg live weight (lbs)	1,400	1,450	975	700		

Predominant milk cow breed: Holstein

Average milk production: 74 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 40,982.87 *tons per reporting period*

Total nitrogen from manure: 496,366.76 *lbs per reporting period*

After ammonia losses (30% loss applied): 347,456.73 *lbs per reporting period*

Total phosphorus from manure: 81,060.01 *lbs per reporting period*

Total potassium from manure: 192,195.85 *lbs per reporting period*

Total salt from manure: 502,933.50 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 15,945,800 *gallons*

Total nitrogen generated: 121,348.95 *lbs*

Total phosphorus generated: 9,719.55 *lbs*

Total potassium generated: 109,833.42 *lbs*

Total salt generated: 928,877.74 *lbs*

	15,945,800	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	15,945,800	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
MID Lateral 5	Surface water
OID	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

Date	Material type / Description	Quantity	Reporting basis	Moist. (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
01/01/2023	Corral solids *Corral Manure - 11/23/2022*	710.00 *ton*	Dry-weight	65.7	23,700.00	12,900.00	77,800.00		30.00
05/29/2023	Corral solids *Corral Manure*	520.00 *ton*	Dry-weight	65.8	23,700.00	6,860.00	28,400.00		36.40

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
03/01/2023	Solid commercial fertilizer *CAN17*	27.72 *ton*	Dry-weight	0.1	17.000000	0.000000	0.000000	0.000000
06/05/2023	Solid commercial fertilizer *UN32 - Almonds*	13.36 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000
07/11/2023	Solid commercial fertilizer *UN32 - Corn*	8.85 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	23,615.56	0.00	0.00	0.00
Dry manure	19,972.94	8,723.04	47,994.58	275,585.52
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	43,588.50	8,723.04	47,994.58	275,585.52

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	50	50	2	process wastewater	X014-X013-X003-XXXX
Field 2	46	46	2	process wastewater	X014-X013-X003-XXXX
Field 3	35	35	2	process wastewater	X014-X013-X003-XXXX
Field 4	55	55	2	process wastewater	X014-X013-X003-XXXX
Field 5	36	36	2	process wastewater	X014-X013-X003-XXXX
Field 6	30	30	2	process wastewater	X014-X013-X003-XXXX
Field B1	55	55	1	none	X014-X010-X005-XXXX X014-X011-X001-XXXX
Field B2	72	72	2	manure	X014-X010-X005-XXXX X014-X011-X001-XXXX
Field B3	182	182	1	none	X014-X010-X005-XXXX X014-X010-X006-XXXX
Field B4	85	85	1	none	X014-X010-X006-XXXX
Field B5	131	131	1	manure	X014-X010-X005-XXXX X014-X010-X006-XXXX X014-X011-X001-XXXX
Squirrel Field 1	44	44	1	none	X014-X024-X009-XXXX
Squirrel Field 2	89	89	2	manure	X014-X023-X007-XXXX
Totals for areas that were used for application	544	544	17		
Totals for areas that were not used for application	366	366	4		
Land application area totals	910	910	21		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

Field 1

04/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 50 Plant date: 04/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/18/2023	1,140.00 *ton*	As-is		53.6	5,990.00	870.00	6,010.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	22.80	273.14	39.67	274.06	1,396.45

08/20/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 50 Plant date: 08/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/15/2023	377.00 *ton*	As-is		70.0	5,500.00	900.00	6,000.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	7.54	82.94	13.57	90.48	0.00

Field 2

Field name: Field 2

04/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 46 Plant date: 04/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/18/2023	1,520.00 *ton*	As-is		64.0	4,100.00	650.00	6,120.00		7.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	33.04	270.96	42.96	404.45	1,736.77

Field 2

08/20/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 46 Plant date: 08/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/15/2023	354.00 *ton*	As-is		70.0	5,500.00	900.00	6,000.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	7.70	84.65	13.85	92.35	0.00

Field 3

Field name: Field 3

11/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 11/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/11/2023	657.00 *ton*	As-is		74.6	3,530.00	640.00	5,640.00		9.37

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.77	132.53	24.03	211.74	893.51

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/22/2023	812.00 *ton*	As-is		63.8	4,630.00	600.00	5,070.00		11.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	23.20	214.83	27.84	235.25	1,847.65

Field 4

Field name: Field 4

11/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 55 Plant date: 11/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/12/2023	1,354.00 *ton*	As-is		81.4	3,480.00	570.00	4,880.00		9.92

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	24.62	171.34	28.06	240.27	908.47

06/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 55 Plant date: 06/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/17/2023	1,362.00 *ton*	As-is		60.6	5,240.00	590.00	4,270.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	24.76	259.52	29.22	211.48	1,287.91

Field 5

Field name: Field 5

11/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 36 Plant date: 11/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/11/2023	662.00 *ton*	As-is		75.2	2,750.00	530.00	4,720.00		8.57

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.39	101.14	19.49	173.59	781.66

Field 5

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 36 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	896.00 *ton*	As-is		65.2	4,630.00	380.00	6,460.00		6.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	24.89	230.47	18.92	321.56	1,056.68

Field 6

Field name: Field 6

11/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 11/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/11/2023	590.00 *ton*	As-is		74.6	3,530.00	640.00	5,640.00		9.37

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	19.67	138.85	25.17	221.84	936.13

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/21/2023	948.00 *ton*	As-is		69.6	4,100.00	560.00	4,960.00		6.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	31.60	259.12	35.39	313.47	1,287.26

Field B1

Field name: Field B1

02/10/2013: Almond, in shell

Crop: Almond, in shell Acres planted: 55 Plant date: 02/10/2013

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/15/2023	341.20 *ton*	As-is		17.2	13,800.00	1,180.00	9,300.00		11.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	258.00	43.80	279.00	0.00
Total actual harvest content	6.20	171.22	14.64	115.39	1,160.87

Field B2

Field name: Field B2

10/31/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 72 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/18/2023	1,061.00 *ton*	As-is		52.6	5,200.00	860.00	6,550.00		6.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.74	153.26	25.35	193.04	963.92

06/19/2023: Corn, silage

Crop: Corn, silage Acres planted: 72 Plant date: 06/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/16/2023	1,786.00 *ton*	As-is		67.0	4,490.00	370.00	5,030.00		6.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	24.81	222.75	18.36	249.54	1,096.90

Field B3

Field name: Field B3

01/01/2023: Pasture

Crop: Pasture Acres planted: 182 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	3,640.00 *ton*	As-is		70.0	3,150.00	340.00	3,400.00		4.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	368.00	52.00	378.00	0.00
Total actual harvest content	20.00	126.00	13.60	136.00	576.00

Field B4

Field name: Field B4

01/01/2023: Pasture

Crop: Pasture Acres planted: 85 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	1,700.00 *ton*	As-is		70.0	3,000.00	390.00	5,200.00		7.91

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	368.00	52.00	378.00	0.00
Total actual harvest content	20.00	120.00	15.60	208.00	949.20

Field B5

Field name: Field B5

Field B5

01/01/2023: Pasture

Crop: Pasture Acres planted: 131 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	2,620.00 *ton*	As-is		70.0	4,020.00	600.00	5,350.00		7.66

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	20.00	368.00	52.00	378.00	0.00
Total actual harvest content	20.00	160.80	24.00	214.00	919.20

Squirrel Field 1

Field name: Squirrel Field 1

02/05/2010: Almond, in shell

Crop: Almond, in shell Acres planted: 44 Plant date: 02/05/2010

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	281.60 *ton*	As-is		31.8	14,300.00	1,540.00	13,300.00		11.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	258.00	43.80	279.00	0.00
Total actual harvest content	6.40	183.04	19.71	170.24	986.44

Squirrel Field 2

Field name: Squirrel Field 2

Squirrel Field 2

11/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 89 Plant date: 11/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/18/2023	1,557.00 *ton*	As-is		67.0	2,730.00	560.00	3,920.00		8.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.49	95.52	19.59	137.16	981.43

06/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 89 Plant date: 06/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/16/2023	2,221.00 *ton*	As-is		68.4	4,930.00	480.00	2,860.00		6.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	24.96	246.06	23.96	142.74	946.30

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 04/14/2023: Corn, silage

Field name: Field 1

Crop: Corn, silage

Plant date: 04/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	41.68	6,750,000.00 *gal*
Application event totals		0.00	0.00	0.00	41.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	63.09	6.42	77.47	560.22	504,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	36.68	5,940,000.00 *gal*
Application event totals		63.09	6.42	77.47	596.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	63.09	6.42	77.47	560.22	504,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	36.68	5,940,000.00 *gal*
Application event totals		63.09	6.42	77.47	596.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	63.09	6.42	77.47	560.22	504,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	35.85	5,805,000.00 *gal*
Application event totals		63.09	6.42	77.47	596.07	

Field 1 - 04/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	63.09	6.42	77.47	560.22	504,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	35.01	5,670,000.00 *gal*
Application event totals		63.09	6.42	77.47	595.24	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	63.09	6.42	77.47	560.22	504,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	35.01	5,670,000.00 *gal*
Application event totals		63.09	6.42	77.47	595.24	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	36.68	5,940,000.00 *gal*
Application event totals		0.00	0.00	0.00	36.68	

Field 1 - 08/20/2023: Sudangrass, silage

Field name: Field 1

Crop: Sudangrass, silage

Plant date: 08/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	40.02	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	40.02	

Field 1 - 08/20/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	51.59	2.37	70.80	629.48	420,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	35.85	5,805,000.00 *gal*
Application event totals		51.59	2.37	70.80	665.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	35.01	5,670,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	51.82	3.74	49.88	435.73	504,000.00 *gal*
Application event totals		51.82	3.74	49.88	435.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	36.68	5,940,000.00 *gal*
Application event totals		0.00	0.00	0.00	36.68	

Field 2 - 04/14/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 04/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	41.68	6,210,000.00 *gal*
Application event totals		0.00	0.00	0.00	41.68	

Field 2 - 04/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	68.57	6.98	84.21	608.94	504,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	36.25	5,400,000.00 *gal*
Application event totals		68.57	6.98	84.21	645.18	

06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	68.57	6.98	84.21	608.94	504,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	35.34	5,265,000.00 *gal*
Application event totals		68.57	6.98	84.21	644.28	

06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	68.57	6.98	84.21	608.94	504,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	35.34	5,265,000.00 *gal*
Application event totals		68.57	6.98	84.21	644.28	

07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	68.57	6.98	84.21	608.94	504,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	35.34	5,265,000.00 *gal*
Application event totals		68.57	6.98	84.21	644.28	

07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	68.57	6.98	84.21	608.94	504,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	34.43	5,130,000.00 *gal*
Application event totals		68.57	6.98	84.21	643.37	

Field 2 - 04/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	37.15	5,535,000.00 *gal*
Application event totals		0.00	0.00	0.00	37.15	

Field 2 - 08/20/2023: Sudangrass, silage

Field name: Field 2

Crop: Sudangrass, silage

Plant date: 08/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	39.87	5,940,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	61.69	2.83	84.65	752.64	462,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	36.25	5,400,000.00 *gal*
Application event totals		61.69	2.83	84.65	788.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	46.94	3.39	45.18	394.68	420,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	34.43	5,130,000.00 *gal*
Application event totals		46.94	3.39	45.18	429.12	

Field 2 - 08/20/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	36.25	5,400,000.00 *gal*
Application event totals		0.00	0.00	0.00	36.25	

Field 3 - 11/30/2022: Oats, silage-soft dough

Field name: Field 3

Crop: Oats, silage-soft dough Plant date: 11/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/21/2022	Towed tank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater sludge	102.34	7.68	62.14	216.87	255,500.00 *gal*
Application event totals		102.34	7.68	62.14	216.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	72.48	6.95	58.60	608.05	462,000.00 *gal*
Application event totals		72.48	6.95	58.60	608.05	

Field 3 - 06/12/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 3 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Towed tank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater sludge	143.16	14.13	1.70	403.28	255,500.00 *gal*
Application event totals		143.16	14.13	1.70	403.28	

05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	42.87	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	42.87	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	40.49	4,590,000.00 *gal*
Application event totals		0.00	0.00	0.00	40.49	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	81.07	3.72	111.26	989.18	462,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	35.73	4,050,000.00 *gal*
Application event totals		81.07	3.72	111.26	1,024.91	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	39.30	4,455,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.30	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	66.33	3.05	91.03	809.33	378,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	35.73	4,050,000.00 *gal*
Application event totals		66.33	3.05	91.03	845.06	

Field 3 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	39.30	4,455,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.30	

Field 4 - 11/30/2022: Oats, silage-soft dough

Field name: Field 4

Crop: Oats, silage-soft dough

Plant date: 11/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	102.34	7.68	62.14	216.87	401,500.00 *gal*
Application event totals		102.34	7.68	62.14	216.87	

11/20/2023	Towed tank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	117.41	11.26	94.93	984.94	1,176,000.00 *gal*
Application event totals		117.41	11.26	94.93	984.94	

Field 4 - 06/22/2023: Corn, silage

Field name: Field 4

Crop: Corn, silage

Plant date: 06/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 4 - 06/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Towed tank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	143.16	14.13	1.70	403.28	401,500.00 *gal*
Application event totals		143.16	14.13	1.70	403.28	

06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	42.44	7,560,000.00 *gal*
Application event totals		0.00	0.00	0.00	42.44	

07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	40.17	7,155,000.00 *gal*
Application event totals		0.00	0.00	0.00	40.17	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.66	3.02	90.11	801.16	588,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	35.62	6,345,000.00 *gal*
Application event totals		65.66	3.02	90.11	836.78	

08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.66	3.02	90.11	801.16	588,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	35.62	6,345,000.00 *gal*
Application event totals		65.66	3.02	90.11	836.78	

08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	39.41	7,020,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.41	

Field 4 - 06/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.66	3.02	90.11	801.16	588,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	34.86	6,210,000.00 *gal*
Application event totals		65.66	3.02	90.11	836.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	37.89	6,750,000.00 *gal*
Application event totals		0.00	0.00	0.00	37.89	

Field 5 - 11/30/2022: Oats, silage-soft dough

Field name: Field 5

Crop: Oats, silage-soft dough Plant date: 11/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/20/2023	Towed tank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	142.14	10.66	86.30	301.21	365,000.00 *gal*
Application event totals		142.14	10.66	86.30	301.21	

Field 5 - 06/20/2023: Corn, silage

Field name: Field 5

Crop: Corn, silage Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 5 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Towed tank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	143.16	14.13	1.70	403.28	262,800.00 *gal*
Application event totals		143.16	14.13	1.70	403.28	

06/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	42.26	4,927,500.00 *gal*
Application event totals		0.00	0.00	0.00	42.26	

07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	40.53	4,725,000.00 *gal*
Application event totals		0.00	0.00	0.00	40.53	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	57.32	2.63	78.67	699.42	336,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	38.21	4,455,000.00 *gal*
Application event totals		57.32	2.63	78.67	737.63	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	57.32	2.63	78.67	699.42	336,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	38.21	4,455,000.00 *gal*
Application event totals		57.32	2.63	78.67	737.63	

08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	39.37	4,590,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.37	

Field 5 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	57.32	2.63	78.67	699.42	336,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	35.89	4,185,000.00 *gal*
Application event totals		57.32	2.63	78.67	735.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	38.21	4,455,000.00 *gal*
Application event totals		0.00	0.00	0.00	38.21	

Field 6 - 11/30/2022: Oats, silage-soft dough

Field name: Field 6

Crop: Oats, silage-soft dough Plant date: 11/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	84.56	8.11	68.37	709.39	462,000.00 *gal*
Application event totals		84.56	8.11	68.37	709.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/20/2023	Towed tank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	102.34	7.68	62.14	216.87	219,000.00 *gal*
Application event totals		102.34	7.68	62.14	216.87	

Field 6 - 06/12/2023: Corn, silage

Field name: Field 6

Crop: Corn, silage Plant date: 06/12/2023

Field 6 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Towed tank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	143.16	14.13	1.70	403.28	219,000.00 *gal*
Application event totals		143.16	14.13	1.70	403.28	

05/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	42.38	4,117,500.00 *gal*
Application event totals		0.00	0.00	0.00	42.38	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	40.29	3,915,000.00 *gal*
Application event totals		0.00	0.00	0.00	40.29	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	68.79	3.16	94.40	839.31	336,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	38.21	3,712,500.00 *gal*
Application event totals		68.79	3.16	94.40	877.52	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	68.79	3.16	94.40	839.31	336,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	36.13	3,510,000.00 *gal*
Application event totals		68.79	3.16	94.40	875.43	

08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 5	Surface water	0.00	0.00	0.00	38.90	3,780,000.00 *gal*
Application event totals		0.00	0.00	0.00	38.90	

Field 6 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	68.79	3.16	94.40	839.31	336,000.00 *gal*
MID Lateral 5	Surface water	0.00	0.00	0.00	36.13	3,510,000.00 *gal*
Application event totals		68.79	3.16	94.40	875.43	

Field B1 - 02/10/2013: Almond, in shell

Field name: Field B1

Crop: Almond, in shell

Plant date: 02/10/2013

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/29/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.00	0.00	0.00	0.00	3,410,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/07/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.00	0.00	0.00	0.00	3,410,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/14/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.00	0.00	0.00	0.00	3,410,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/03/2022	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	44.20	0.00	0.00	0.00	
Application event totals		44.20	0.00	0.00	0.00	

Field B1 - 02/10/2013: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/01/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	51.00	0.00	0.00	0.00	
Application event totals		51.00	0.00	0.00	0.00	

04/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.82	0.00	0.00	42.09	5,335,000.00 *gal*
Application event totals		0.82	0.00	0.00	42.09	

04/21/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	43.20	0.00	0.00	0.00	
Application event totals		43.20	0.00	0.00	0.00	

04/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.82	0.00	0.00	42.09	5,335,000.00 *gal*
Application event totals		0.82	0.00	0.00	42.09	

05/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.82	0.00	0.00	42.09	5,335,000.00 *gal*
Application event totals		0.82	0.00	0.00	42.09	

05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.82	0.00	0.00	42.09	5,335,000.00 *gal*
Application event totals		0.82	0.00	0.00	42.09	

Field B1 - 02/10/2013: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	43.20	0.00	0.00	0.00	
Application event totals		43.20	0.00	0.00	0.00	

06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.82	0.00	0.00	42.09	5,335,000.00 *gal*
Application event totals		0.82	0.00	0.00	42.09	

06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.83	0.00	0.00	42.63	5,403,750.00 *gal*
Application event totals		0.83	0.00	0.00	42.63	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.83	0.00	0.00	42.63	5,403,750.00 *gal*
Application event totals		0.83	0.00	0.00	42.63	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.83	0.00	0.00	42.63	5,403,750.00 *gal*
Application event totals		0.83	0.00	0.00	42.63	

08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.83	0.00	0.00	42.63	5,403,750.00 *gal*
Application event totals		0.83	0.00	0.00	42.63	

Field B1 - 02/10/2013: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.83	0.00	0.00	42.63	5,403,750.00 *gal*
Application event totals		0.83	0.00	0.00	42.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.83	0.00	0.00	42.63	5,403,750.00 *gal*
Application event totals		0.83	0.00	0.00	42.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.83	0.00	0.00	42.63	5,403,750.00 *gal*
Application event totals		0.83	0.00	0.00	42.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.83	0.00	0.00	42.63	5,403,750.00 *gal*
Application event totals		0.83	0.00	0.00	42.63	

Field B2 - 10/31/2022: Oats, silage-soft dough

Field name: Field B2

Crop: Oats, silage-soft dough

Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Heifer Corral Manure	Corral solids	206.89	94.68	475.15	3,366.40	600.00 *ton*
Application event totals		206.89	94.68	475.15	3,366.40	

Field B2 - 06/19/2023: Corn, silage

Field name: Field B2

Crop: Corn, silage

Plant date: 06/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Heifer Corral Manure	Corral solids	248.24	74.10	255.59	11,124.61	1,340.00 *ton*
Application event totals		248.24	74.10	255.59	11,124.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	1.14	0.00	0.00	58.76	9,750,000.00 *gal*
Application event totals		1.14	0.00	0.00	58.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	1.05	0.00	0.00	54.24	9,000,000.00 *gal*
Application event totals		1.05	0.00	0.00	54.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	1.09	0.00	0.00	56.05	9,300,000.00 *gal*
Application event totals		1.09	0.00	0.00	56.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	1.07	0.00	0.00	55.15	9,150,000.00 *gal*
Application event totals		1.07	0.00	0.00	55.15	

Field B2 - 06/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	1.05	0.00	0.00	54.24	9,000,000.00 *gal*
Application event totals		1.05	0.00	0.00	54.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	1.04	0.00	0.00	53.34	8,850,000.00 *gal*
Application event totals		1.04	0.00	0.00	53.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	1.02	0.00	0.00	52.43	8,700,000.00 *gal*
Application event totals		1.02	0.00	0.00	52.43	

Field B3 - 01/01/2023: Pasture

Field name: Field B3

Crop: Pasture

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.84	0.00	0.00	43.45	18,225,000.00 *gal*
Application event totals		0.84	0.00	0.00	43.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.81	0.00	0.00	41.72	17,496,000.00 *gal*
Application event totals		0.81	0.00	0.00	41.72	

Field B3 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.83	0.00	0.00	42.87	17,982,000.00 *gal*
Application event totals		0.83	0.00	0.00	42.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.84	0.00	0.00	43.45	18,225,000.00 *gal*
Application event totals		0.84	0.00	0.00	43.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.86	0.00	0.00	44.03	18,468,000.00 *gal*
Application event totals		0.86	0.00	0.00	44.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.86	0.00	0.00	44.03	18,468,000.00 *gal*
Application event totals		0.86	0.00	0.00	44.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.87	0.00	0.00	44.61	18,711,000.00 *gal*
Application event totals		0.87	0.00	0.00	44.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.84	0.00	0.00	43.16	18,103,500.00 *gal*
Application event totals		0.84	0.00	0.00	43.16	

Field B3 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.83	0.00	0.00	42.87	17,982,000.00 *gal*
Application event totals		0.83	0.00	0.00	42.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.83	0.00	0.00	42.87	17,982,000.00 *gal*
Application event totals		0.83	0.00	0.00	42.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.81	0.00	0.00	41.72	17,496,000.00 *gal*
Application event totals		0.81	0.00	0.00	41.72	

Field B4 - 01/01/2023: Pasture

Field name: Field B4

Crop: Pasture

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.86	0.00	0.00	44.11	8,640,000.00 *gal*
Application event totals		0.86	0.00	0.00	44.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.84	0.00	0.00	43.19	8,460,000.00 *gal*
Application event totals		0.84	0.00	0.00	43.19	

Field B4 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.85	0.00	0.00	43.65	8,550,000.00 *gal*
Application event totals		0.85	0.00	0.00	43.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.86	0.00	0.00	44.11	8,640,000.00 *gal*
Application event totals		0.86	0.00	0.00	44.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.87	0.00	0.00	45.03	8,820,000.00 *gal*
Application event totals		0.87	0.00	0.00	45.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.87	0.00	0.00	45.03	8,820,000.00 *gal*
Application event totals		0.87	0.00	0.00	45.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.87	0.00	0.00	44.57	8,730,000.00 *gal*
Application event totals		0.87	0.00	0.00	44.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.86	0.00	0.00	44.11	8,640,000.00 *gal*
Application event totals		0.86	0.00	0.00	44.11	

Field B4 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.84	0.00	0.00	43.19	8,460,000.00 *gal*
Application event totals		0.84	0.00	0.00	43.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.84	0.00	0.00	43.19	8,460,000.00 *gal*
Application event totals		0.84	0.00	0.00	43.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.82	0.00	0.00	42.27	8,280,000.00 *gal*
Application event totals		0.82	0.00	0.00	42.27	

Field B5 - 01/01/2023: Pasture

Field name: Field B5

Crop: Pasture

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.85	0.00	0.00	43.82	13,230,000.00 *gal*
Application event totals		0.85	0.00	0.00	43.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.84	0.00	0.00	43.23	13,050,000.00 *gal*
Application event totals		0.84	0.00	0.00	43.23	

Field B5 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Heifer Corral Manure	Corral solids	171.05	51.06	176.12	7,665.67	1,680.00 *ton*
Application event totals		171.05	51.06	176.12	7,665.67	

05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.83	0.00	0.00	42.93	12,960,000.00 *gal*
Application event totals		0.83	0.00	0.00	42.93	

06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.86	0.00	0.00	44.12	13,320,000.00 *gal*
Application event totals		0.86	0.00	0.00	44.12	

06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.85	0.00	0.00	43.82	13,230,000.00 *gal*
Application event totals		0.85	0.00	0.00	43.82	

07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.86	0.00	0.00	44.42	13,410,000.00 *gal*
Application event totals		0.86	0.00	0.00	44.42	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.86	0.00	0.00	44.42	13,410,000.00 *gal*
Application event totals		0.86	0.00	0.00	44.42	

Field B5 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.85	0.00	0.00	43.82	13,230,000.00 *gal*
Application event totals		0.85	0.00	0.00	43.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.84	0.00	0.00	43.23	13,050,000.00 *gal*
Application event totals		0.84	0.00	0.00	43.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.83	0.00	0.00	42.93	12,960,000.00 *gal*
Application event totals		0.83	0.00	0.00	42.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.82	0.00	0.00	42.33	12,780,000.00 *gal*
Application event totals		0.82	0.00	0.00	42.33	

Squirrel Field 1 - 02/05/2010: Almond, in shell

Field name: Squirrel Field 1

Crop: Almond, in shell

Plant date: 02/05/2010

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.00	0.00	0.00	0.00	4,514,400.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Squirrel Field 1 - 02/05/2010: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.00	0.00	0.00	0.00	4,514,400.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/02/2022	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	44.20	0.00	0.00	0.00	
Application event totals		44.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/01/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	51.00	0.00	0.00	0.00	
Application event totals		51.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	43.20	0.00	0.00	0.00	
Application event totals		43.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.83	0.00	0.00	42.53	4,312,000.00 *gal*
Application event totals		0.83	0.00	0.00	42.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.83	0.00	0.00	42.53	4,312,000.00 *gal*
Application event totals		0.83	0.00	0.00	42.53	

Squirrel Field 1 - 02/05/2010: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.83	0.00	0.00	42.53	4,312,000.00 *gal*
Application event totals		0.83	0.00	0.00	42.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	43.20	0.00	0.00	0.00	
Application event totals		43.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.83	0.00	0.00	42.53	4,312,000.00 *gal*
Application event totals		0.83	0.00	0.00	42.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.84	0.00	0.00	43.18	4,378,000.00 *gal*
Application event totals		0.84	0.00	0.00	43.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.84	0.00	0.00	43.18	4,378,000.00 *gal*
Application event totals		0.84	0.00	0.00	43.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.84	0.00	0.00	43.18	4,378,000.00 *gal*
Application event totals		0.84	0.00	0.00	43.18	

Squirrel Field 1 - 02/05/2010: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.84	0.00	0.00	43.18	4,378,000.00 *gal*
Application event totals		0.84	0.00	0.00	43.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.84	0.00	0.00	43.18	4,378,000.00 *gal*
Application event totals		0.84	0.00	0.00	43.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.84	0.00	0.00	43.18	4,378,000.00 *gal*
Application event totals		0.84	0.00	0.00	43.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.84	0.00	0.00	43.18	4,378,000.00 *gal*
Application event totals		0.84	0.00	0.00	43.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	0.84	0.00	0.00	43.18	4,378,000.00 *gal*
Application event totals		0.84	0.00	0.00	43.18	

Squirrel Field 2 - 11/30/2022: Oats, silage-soft dough

Field name: Squirrel Field 2

Crop: Oats, silage-soft dough

Plant date: 11/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Squirrel Field 2 - 11/30/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/23/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	129.70	70.60	425.77	1,641.78	710.00 *ton*
Application event totals		129.70	70.60	425.77	1,641.78	

11/23/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Manure	Separator solids	38.66	5.85	12.34	131.18	400.00 *ton*
Application event totals		38.66	5.85	12.34	131.18	

Squirrel Field 2 - 06/16/2023: Corn, silage

Field name: Squirrel Field 2

Crop: Corn, silage

Plant date: 06/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separated Solids	Separator solids	126.81	21.28	58.29	593.26	1,500.00 *ton*
Heifer Corral Manure	Corral solids	64.44	19.24	66.35	2,887.96	430.00 *ton*
Corral Manure	Corral solids	94.71	27.42	113.50	1,454.69	520.00 *ton*
Application event totals		285.97	67.94	238.14	4,935.91	

06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	1.11	0.00	0.00	57.05	11,700,000.00 *gal*
Application event totals		1.11	0.00	0.00	57.05	

Squirrel Field 2 - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	1.06	0.00	0.00	54.41	11,160,000.00 *gal*
Application event totals		1.06	0.00	0.00	54.41	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	1.05	0.00	0.00	53.97	11,070,000.00 *gal*
Application event totals		1.05	0.00	0.00	53.97	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	1.04	0.00	0.00	53.54	10,980,000.00 *gal*
Application event totals		1.04	0.00	0.00	53.54	

08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	1.02	0.00	0.00	52.66	10,800,000.00 *gal*
Application event totals		1.02	0.00	0.00	52.66	

09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	1.03	0.00	0.00	53.10	10,890,000.00 *gal*
Application event totals		1.03	0.00	0.00	53.10	

Squirrel Field 2 - 06/16/2023: Corn, silage

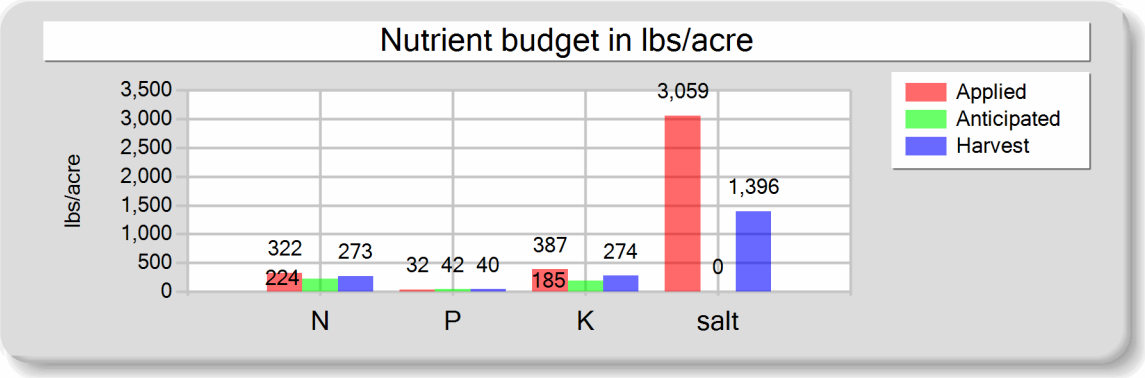
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID	Surface water	1.01	0.00	0.00	51.78	10,620,000.00 *gal*
Application event totals		1.01	0.00	0.00	51.78	

B. NUTRIENT BUDGET

Field 1 - 04/14/2023: Corn, silage

Field name: Field 1 Crop: Corn, silage Plant date: 04/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	315.44	32.09	387.36	2,801.12
Fresh water	0.00	0.00	0.00	257.60
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	322.44	32.09	387.36	3,058.72
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	273.14	39.67	274.06	1,396.45
Nutrient balance	49.30	-7.58	113.31	1,662.26
Applied to removed ratio	1.18	0.81	1.41	2.19

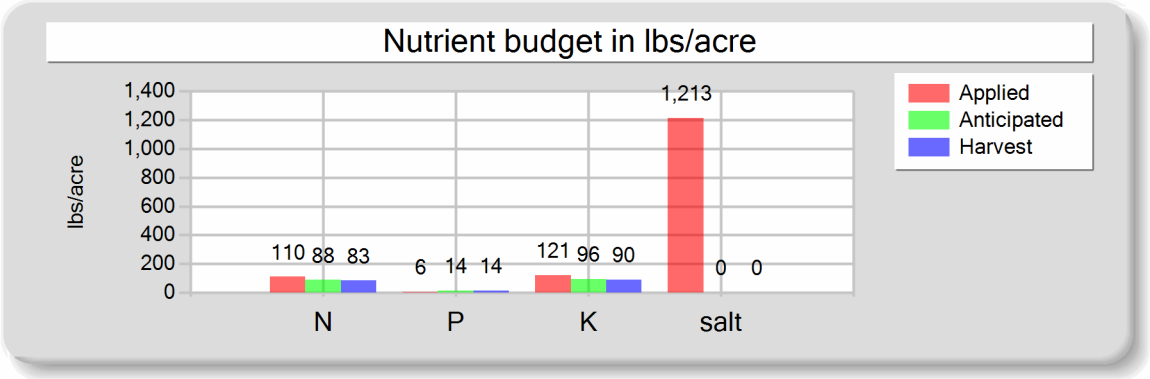
Fresh water applied	
41,715,000.00	*gallons*
1,536.22	*acre-inches*
30.72	*inches/acre*

Process wastewater applied	
2,520,000.00	*gallons*
92.80	*acre-inches*
1.86	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 08/20/2023: Sudangrass, silage

Field name: Field 1 Crop: Sudangrass, silage Plant date: 08/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	103.41	6.11	120.68	1,065.21
Fresh water	0.00	0.00	0.00	147.56
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	110.41	6.11	120.68	1,212.77
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	82.94	13.57	90.48	0.00
Nutrient balance	27.47	-7.46	30.20	1,212.77
Applied to removed ratio	1.33	0.45	1.33	0.00

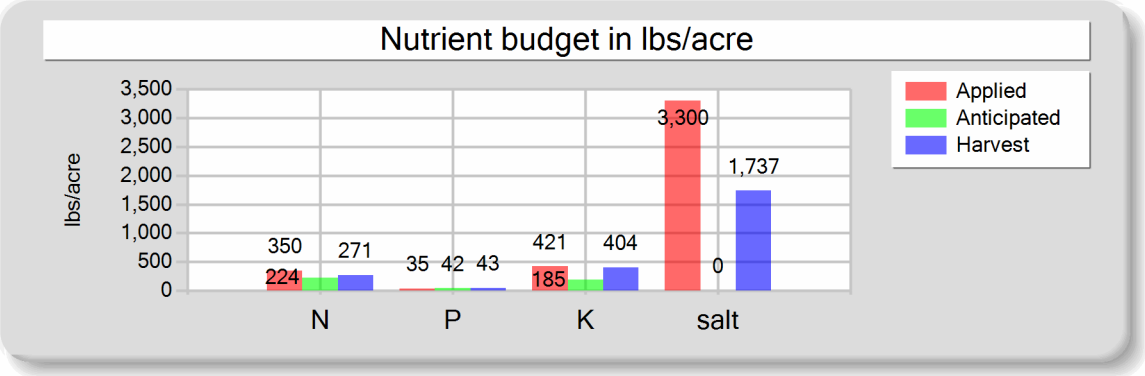
Fresh water applied	
23,895,000.00	*gallons*
879.97	*acre-inches*
17.60	*inches/acre*

Process wastewater applied	
924,000.00	*gallons*
34.03	*acre-inches*
0.68	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 04/14/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 04/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	342.87	34.88	421.05	3,044.69
Fresh water	0.00	0.00	0.00	255.54
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	349.87	34.88	421.05	3,300.23
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	270.96	42.96	404.45	1,736.77
Nutrient balance	78.91	-8.08	16.59	1,563.46
Applied to removed ratio	1.29	0.81	1.04	1.90

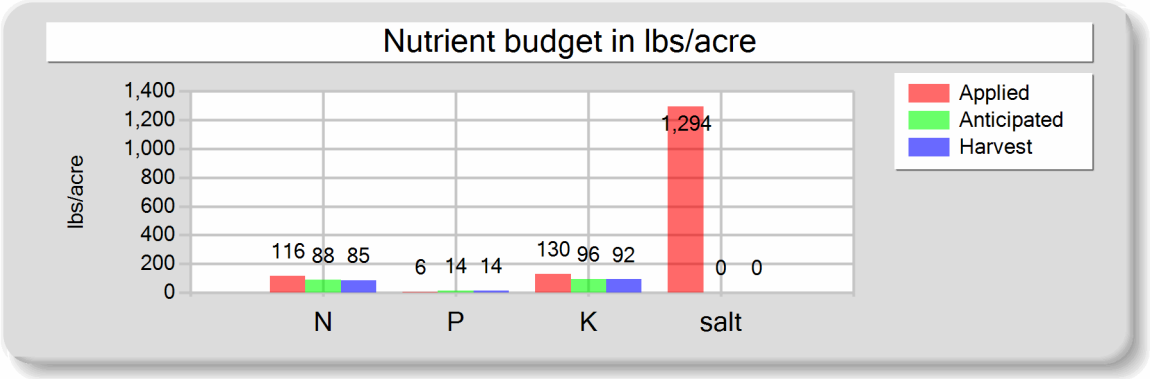
Fresh water applied	
38,070,000.00	*gallons*
1,401.99	*acre-inches*
30.48	*inches/acre*

Process wastewater applied	
2,520,000.00	*gallons*
92.80	*acre-inches*
2.02	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 08/20/2023: Sudangrass, silage

Field name: Field 2 Crop: Sudangrass, silage Plant date: 08/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	108.62	6.22	129.83	1,147.32
Fresh water	0.00	0.00	0.00	146.80
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	115.62	6.22	129.83	1,294.12
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	84.65	13.85	92.35	0.00
Nutrient balance	30.97	-7.63	37.49	1,294.12
Applied to removed ratio	1.37	0.45	1.41	0.00

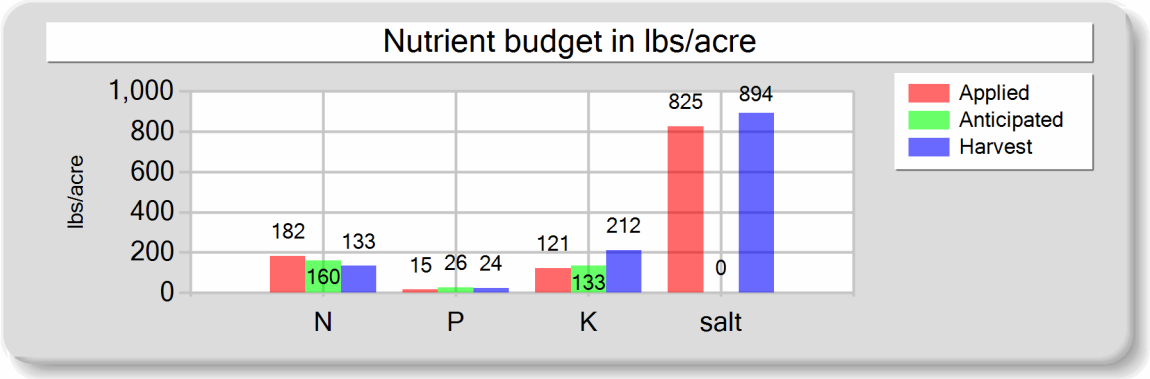
Fresh water applied	
21,870,000.00	*gallons*
805.40	*acre-inches*
17.51	*inches/acre*

Process wastewater applied	
882,000.00	*gallons*
32.48	*acre-inches*
0.71	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 11/30/2022: Oats, silage-soft dough

Field name: Field 3 Crop: Oats, silage-soft dough Plant date: 11/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	174.82	14.63	120.74	824.92
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	181.82	14.63	120.74	824.92
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	132.53	24.03	211.74	893.51
Nutrient balance	49.30	-9.40	-91.00	-68.59
Applied to removed ratio	1.37	0.61	0.57	0.92

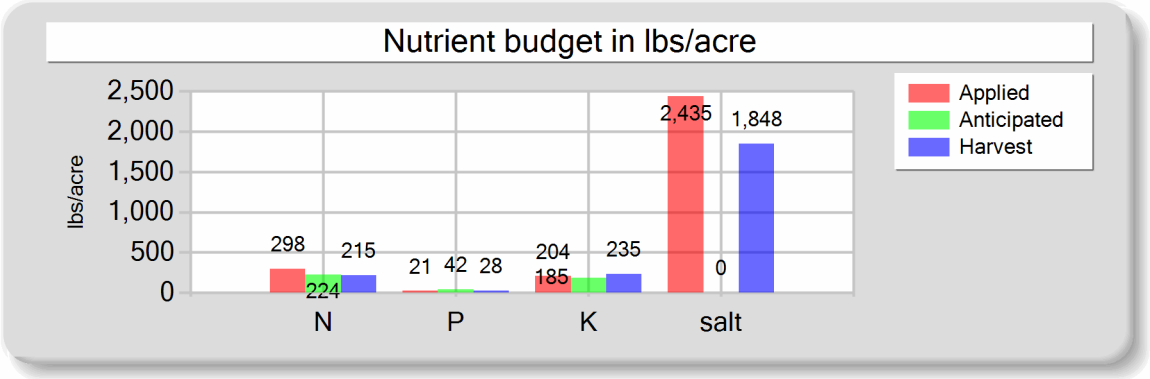
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
717,500.00	*gallons*
26.42	*acre-inches*
0.75	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 06/12/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	290.56	20.90	203.98	2,201.79
Fresh water	0.00	0.00	0.00	233.43
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	297.56	20.90	203.98	2,435.22
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	214.83	27.84	235.25	1,847.65
Nutrient balance	82.73	-6.94	-31.27	587.57
Applied to removed ratio	1.39	0.75	0.87	1.32

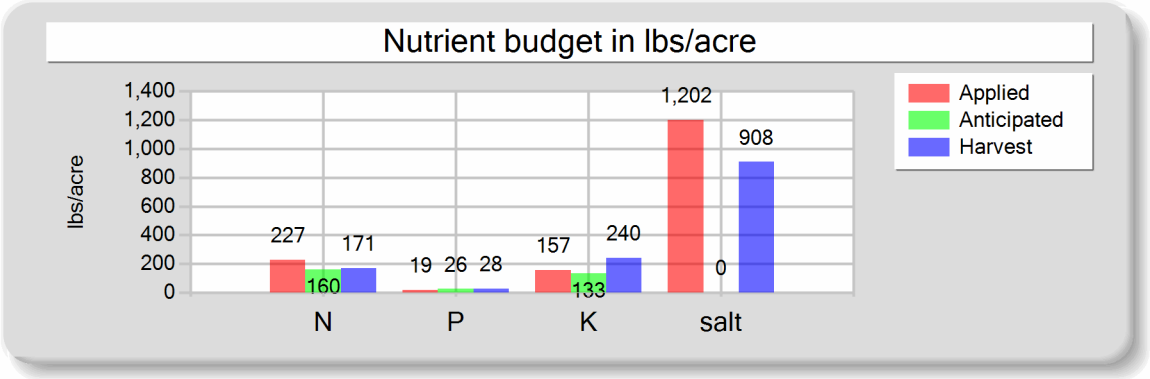
Fresh water applied	
26,460,000.00	*gallons*
974.43	*acre-inches*
27.84	*inches/acre*

Process wastewater applied	
1,095,500.00	*gallons*
40.34	*acre-inches*
1.15	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 11/30/2022: Oats, silage-soft dough

Field name: Field 4 Crop: Oats, silage-soft dough Plant date: 11/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	219.75	18.93	157.06	1,201.81
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	226.75	18.93	157.06	1,201.81
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	171.34	28.06	240.27	908.47
Nutrient balance	55.41	-9.13	-83.21	293.34
Applied to removed ratio	1.32	0.67	0.65	1.32

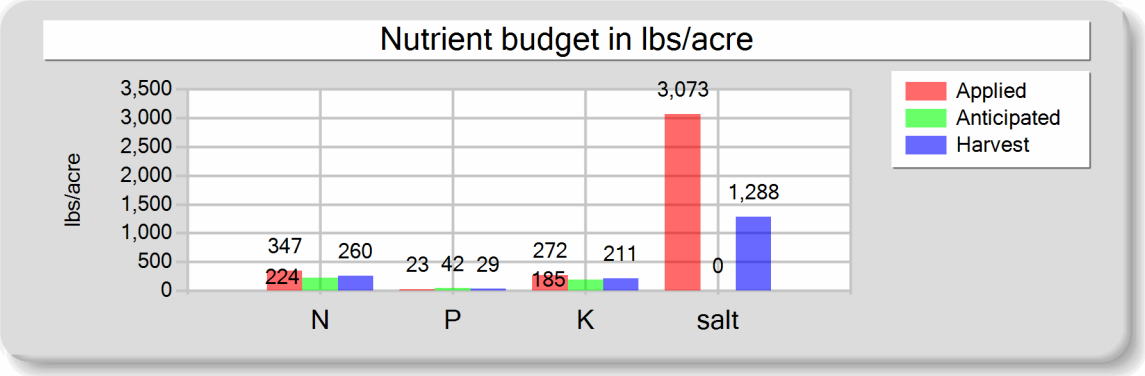
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
1,577,500.00	*gallons*
58.09	*acre-inches*
1.06	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 06/22/2023: Corn, silage

Field name: Field 4 Crop: Corn, silage Plant date: 06/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	340.15	23.18	272.02	2,806.75
Fresh water	0.00	0.00	0.00	266.02
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	347.15	23.18	272.02	3,072.76
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	259.52	29.22	211.48	1,287.91
Nutrient balance	87.62	-6.04	60.54	1,784.86
Applied to removed ratio	1.34	0.79	1.29	2.39

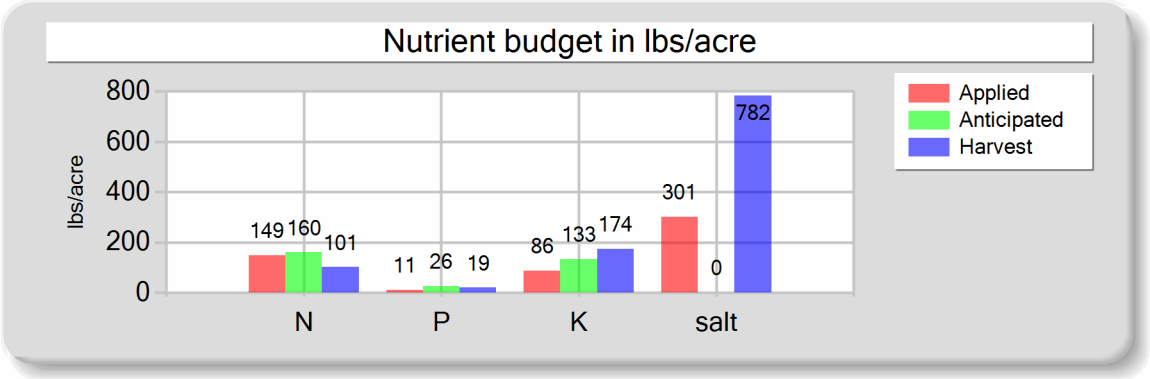
Fresh water applied	
47,385,000.00	*gallons*
1,745.03	*acre-inches*
31.73	*inches/acre*

Process wastewater applied	
2,165,500.00	*gallons*
79.75	*acre-inches*
1.45	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 11/30/2022: Oats, silage-soft dough

Field name: Field 5 Crop: Oats, silage-soft dough Plant date: 11/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	142.14	10.66	86.30	301.21
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	149.14	10.66	86.30	301.21
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	101.14	19.49	173.59	781.66
Nutrient balance	48.00	-8.83	-87.29	-480.45
Applied to removed ratio	1.47	0.55	0.50	0.39

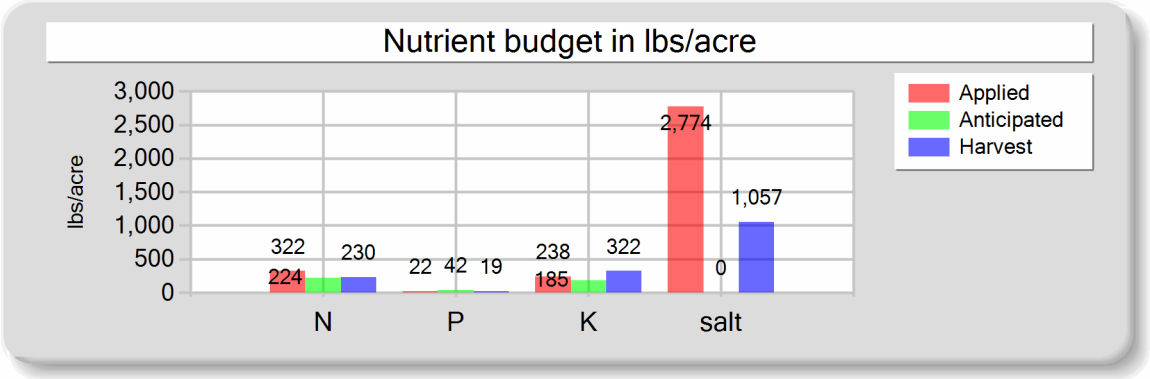
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
365,000.00	*gallons*
13.44	*acre-inches*
0.37	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 06/20/2023: Corn, silage

Field name: Field 5 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	315.13	22.03	237.70	2,501.55
Fresh water	0.00	0.00	0.00	272.68
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	322.13	22.03	237.70	2,774.23
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	230.47	18.92	321.56	1,056.68
Nutrient balance	91.66	3.12	-83.87	1,717.54
Applied to removed ratio	1.40	1.16	0.74	2.63

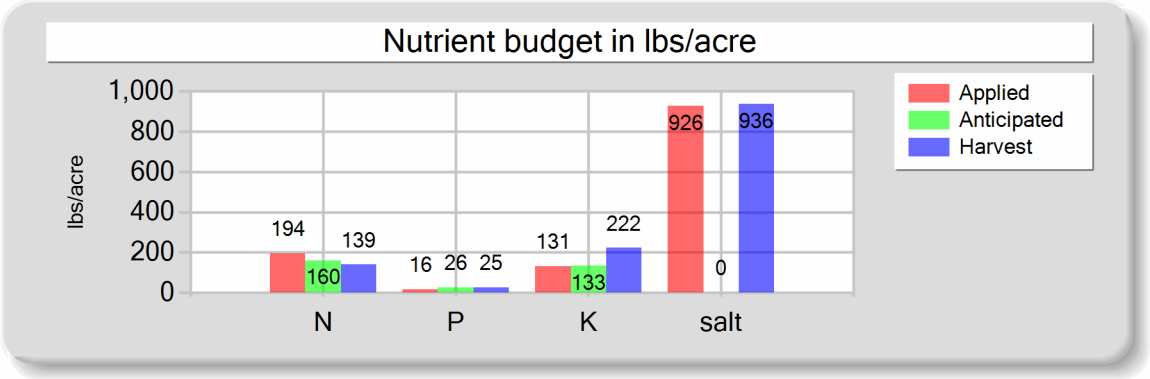
Fresh water applied	
31,792,500.00	*gallons*
1,170.81	*acre-inches*
32.52	*inches/acre*

Process wastewater applied	
1,270,800.00	*gallons*
46.80	*acre-inches*
1.30	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 6 - 11/30/2022: Oats, silage-soft dough

Field name: Field 6 Crop: Oats, silage-soft dough Plant date: 11/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	186.90	15.78	130.51	926.26
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	193.90	15.78	130.51	926.26
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	138.85	25.17	221.84	936.13
Nutrient balance	55.06	-9.39	-91.33	-9.86
Applied to removed ratio	1.40	0.63	0.59	0.99

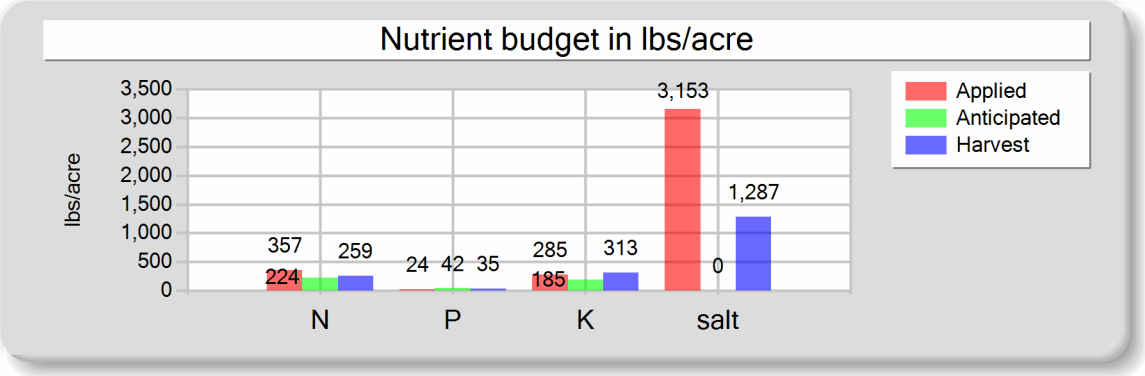
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
681,000.00	*gallons*
25.08	*acre-inches*
0.84	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 6 - 06/12/2023: Corn, silage

Field name: Field 6 Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	349.53	23.61	284.90	2,921.20
Fresh water	0.00	0.00	0.00	232.04
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	356.53	23.61	284.90	3,153.24
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	259.12	35.39	313.47	1,287.26
Nutrient balance	97.41	-11.78	-28.58	1,865.98
Applied to removed ratio	1.38	0.67	0.91	2.45

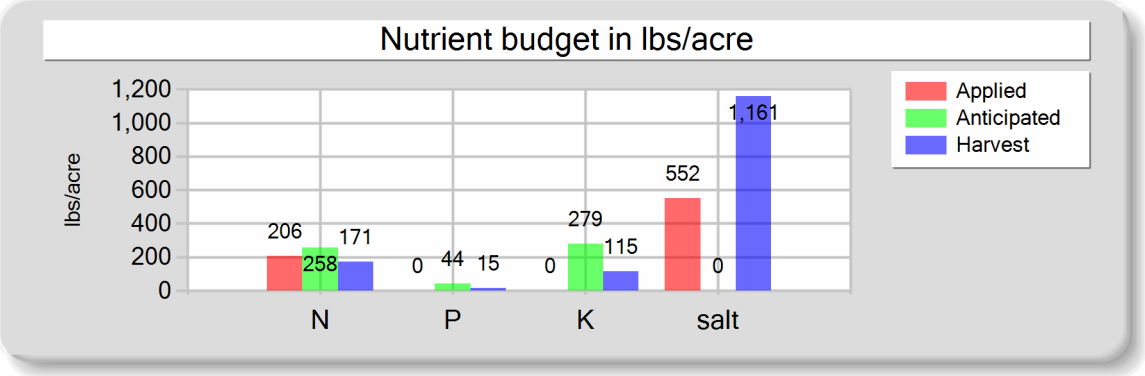
Fresh water applied	
22,545,000.00	*gallons*
830.26	*acre-inches*
27.68	*inches/acre*

Process wastewater applied	
1,227,000.00	*gallons*
45.19	*acre-inches*
1.51	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field B1 - 02/10/2013: Almond, in shell

Field name: Field B1 Crop: Almond, in shell Plant date: 02/10/2013



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	181.60	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	10.71	0.00	0.00	551.54
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	206.31	0.00	0.00	551.54
Anticipated crop nutrient removal	258.00	43.80	279.00	0.00
Actual crop nutrient removal	171.22	14.64	115.39	1,160.87
Nutrient balance	35.09	-14.64	-115.39	-609.34
Applied to removed ratio	1.20	0.00	0.00	0.48

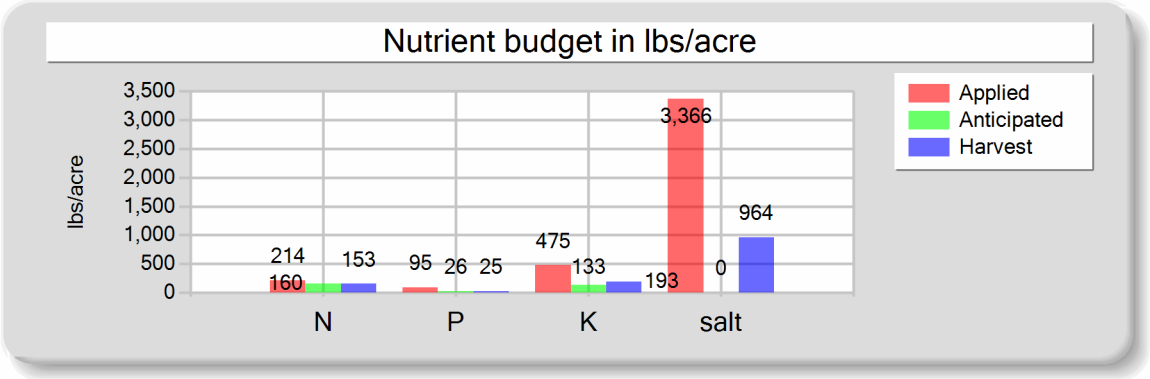
Fresh water applied	
80,135,000.00	*gallons*
2,951.10	*acre-inches*
53.66	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field B2 - 10/31/2022: Oats, silage-soft dough

Field name: Field B2 Crop: Oats, silage-soft dough Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	206.89	94.68	475.15	3,366.40
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	213.89	94.68	475.15	3,366.40
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	153.26	25.35	193.04	963.92
Nutrient balance	60.64	69.33	282.11	2,402.48
Applied to removed ratio	1.40	3.74	2.46	3.49

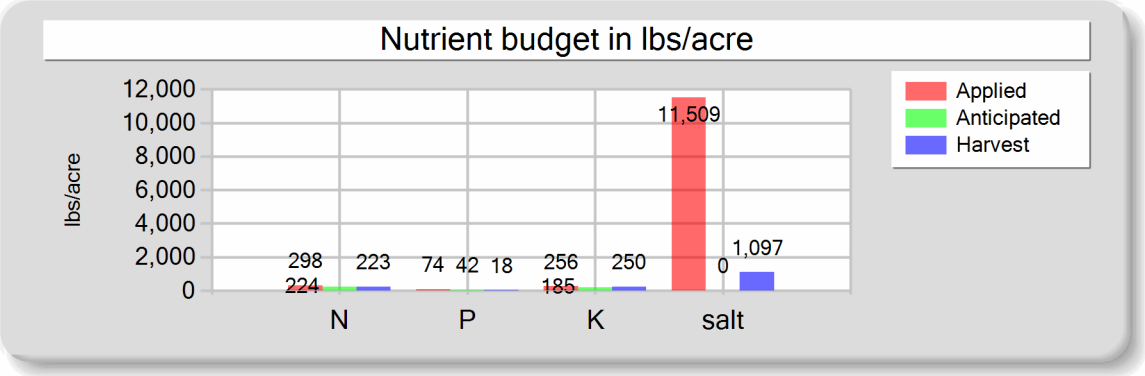
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field B2 - 06/19/2023: Corn, silage

Field name: Field B2 Crop: Corn, silage Plant date: 06/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	35.20	0.00	0.00	0.00
Dry manure	248.24	74.10	255.59	11,124.61
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.46	0.00	0.00	384.22
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	297.90	74.10	255.59	11,508.82
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	222.75	18.36	249.54	1,096.90
Nutrient balance	75.14	55.75	6.05	10,411.92
Applied to removed ratio	1.34	4.04	1.02	10.49

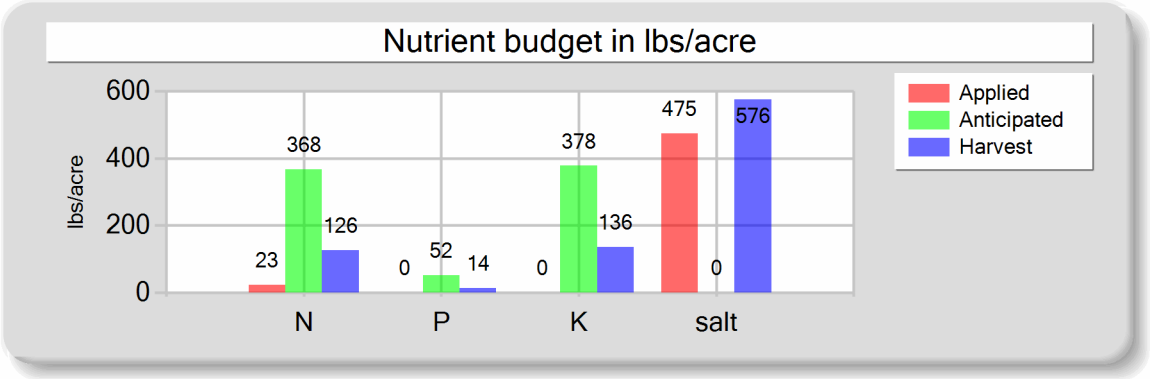
Fresh water applied	
63,750,000.00	*gallons*
2,347.70	*acre-inches*
32.61	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field B3 - 01/01/2023: Pasture

Field name: Field B3 Crop: Pasture Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	9.22	0.00	0.00	474.80
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	23.22	0.00	0.00	474.80
Anticipated crop nutrient removal	368.00	52.00	378.00	0.00
Actual crop nutrient removal	126.00	13.60	136.00	576.00
Nutrient balance	-102.78	-13.60	-136.00	-101.20
Applied to removed ratio	0.18	0.00	0.00	0.82

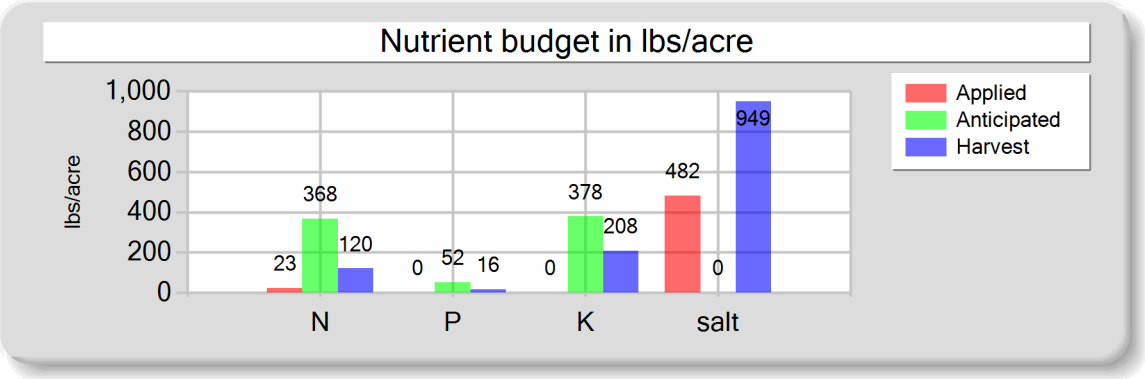
Fresh water applied	
199,138,500.00	*gallons*
7,333.59	*acre-inches*
40.29	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field B4 - 01/01/2023: Pasture

Field name: Field B4 Crop: Pasture Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	9.37	0.00	0.00	482.44
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	23.37	0.00	0.00	482.44
Anticipated crop nutrient removal	368.00	52.00	378.00	0.00
Actual crop nutrient removal	120.00	15.60	208.00	949.20
Nutrient balance	-96.63	-15.60	-208.00	-466.76
Applied to removed ratio	0.19	0.00	0.00	0.51

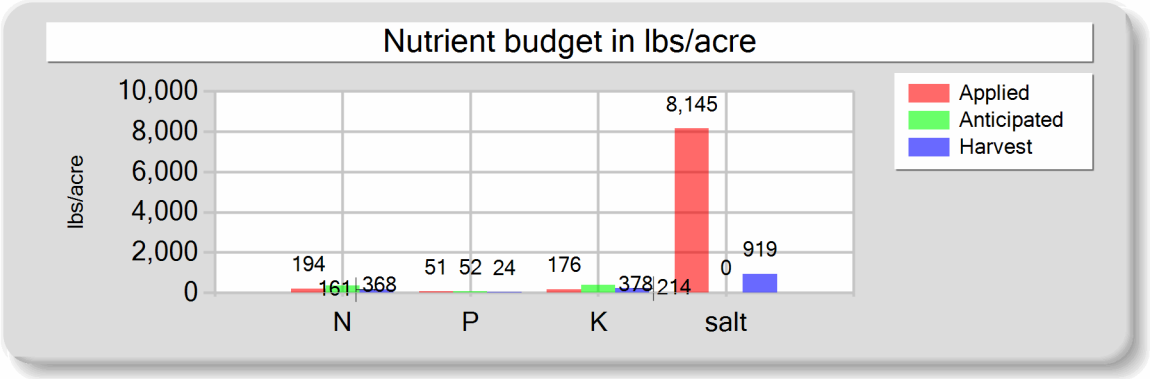
Fresh water applied	
94,500,000.00	*gallons*
3,480.11	*acre-inches*
40.94	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field B5 - 01/01/2023: Pasture

Field name: Field B5 Crop: Pasture Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	171.05	51.06	176.12	7,665.67
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	9.31	0.00	0.00	479.09
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	194.36	51.06	176.12	8,144.76
Anticipated crop nutrient removal	368.00	52.00	378.00	0.00
Actual crop nutrient removal	160.80	24.00	214.00	919.20
Nutrient balance	33.56	27.06	-37.88	7,225.56
Applied to removed ratio	1.21	2.13	0.82	8.86

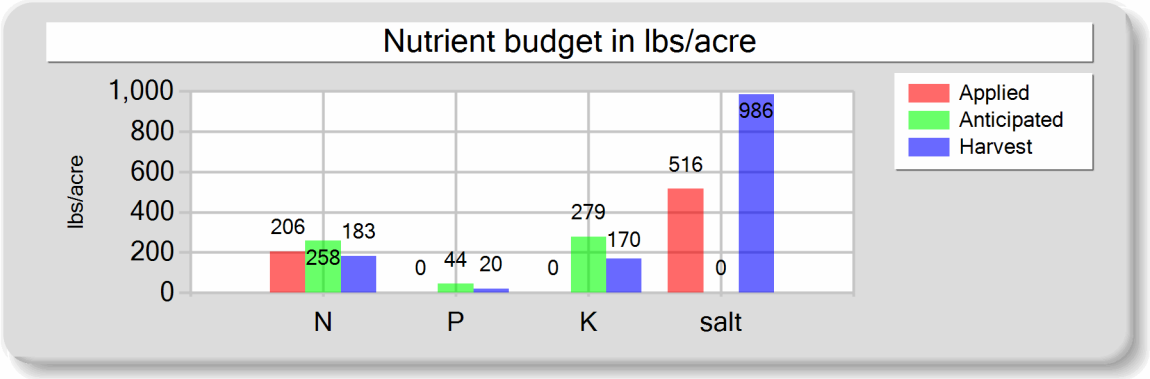
Fresh water applied	
144,630,000.00	*gallons*
5,326.23	*acre-inches*
40.66	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Squirrel Field 1 - 02/05/2010: Almond, in shell

Field name: Squirrel Field 1 Crop: Almond, in shell Plant date: 02/05/2010



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	181.60	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	10.01	0.00	0.00	515.52
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	205.61	0.00	0.00	515.52
Anticipated crop nutrient removal	258.00	43.80	279.00	0.00
Actual crop nutrient removal	183.04	19.71	170.24	986.44
Nutrient balance	22.57	-19.71	-170.24	-470.92
Applied to removed ratio	1.12	0.00	0.00	0.52

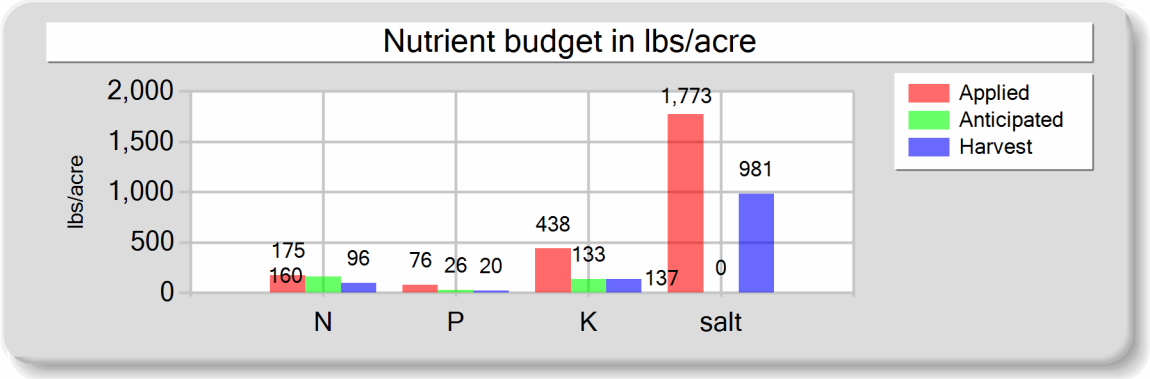
Fresh water applied	
61,300,800.00	*gallons*
2,257.50	*acre-inches*
51.31	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Squirrel Field 2 - 11/30/2022: Oats, silage-soft dough

Field name: Squirrel Field 2 Crop: Oats, silage-soft dough Plant date: 11/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	168.36	76.45	438.10	1,772.95
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	175.36	76.45	438.10	1,772.95
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	95.52	19.59	137.16	981.43
Nutrient balance	79.84	56.85	300.95	791.52
Applied to removed ratio	1.84	3.90	3.19	1.81

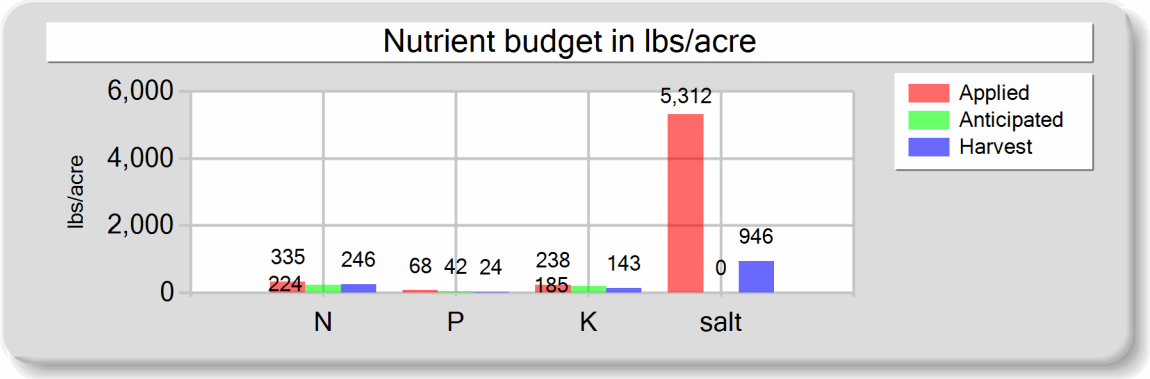
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Squirrel Field 2 - 06/16/2023: Corn, silage

Field name: Squirrel Field 2 Crop: Corn, silage Plant date: 06/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	35.20	0.00	0.00	0.00
Dry manure	285.97	67.94	238.14	4,935.91
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.31	0.00	0.00	376.50
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	335.48	67.94	238.14	5,312.41
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	246.06	23.96	142.74	946.30
Nutrient balance	89.42	43.98	95.39	4,366.12
Applied to removed ratio	1.36	2.84	1.67	5.61

Fresh water applied	
77,220,000.00	*gallons*
2,843.75	*acre-inches*
31.95	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 09/14/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 56.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,300.00	5,390.00	41,800.00	22,000.00	9,520.00	4,910.00	10,600.00	1,540.00		64.40
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 09/14/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,700.00	12,900.00	77,800.00	26,400.00	13,900.00	31,400.00	12,100.00	3,300.00		30.00
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Solids Heifer

Sample and source description: Corral Solids Heifer

Sample date: 09/14/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 47.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,600.00	10,800.00	54,200.00	19,600.00	7,010.00	33,100.00	5,400.00	3,010.00		38.40
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Separated Solids

Sample and source description: Separated Solids

Sample date: 09/14/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,300.00	2,770.00	5,840.00							6.21
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 04/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 44.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,510.00	2,970.00	13,300.00	17,900.00	7,520.00	2,020.00	3,500.00	1,140.00		79.00
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Solids - Imported from Willem Postma #2

Sample and source description: Corral Solids - Imported from Willem Postma #2

Sample date: 04/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,700.00	6,860.00	28,400.00	14,200.00	9,840.00	15,900.00	3,600.00	1,720.00		36.40
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Solids Heifer

Sample and source description: Corral Solids Heifer

Sample date: 04/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 50.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,500.00	4,030.00	13,900.00	13,300.00	5,300.00	6,980.00	2,000.00	1,160.00		60.50
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Separated SOlids

Sample and source description: Separated SOlids

Sample date: 04/19/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 78.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,100.00	2,870.00	7,860.00							8.00
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/11/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 53.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,400.00	5,420.00	20,700.00							64.00
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/11/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 42.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,500.00	3,900.00	22,900.00							65.40
DL	100.00	100.00	100.00							0.01

Separated Solids

Sample and source description: Separated Solids

Sample date: 10/11/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,000.00	2,420.00	6,810.00							5.37
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

Slurry

Sample and source description: Slurry

Sample date: 11/21/2022 Material type: Process wastewater sludge Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,680.00	621.00	0.00	0.00	126.00	1,020.00								7,110.00	3,560
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ1

Sample and source description: WWQ1

Sample date: 02/23/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	658.00	63.10	0.00	0.00	63.10	532.00	162.00	107.00	321.00	3,220.00	0.00	184.00	608.00	8,090.00	5,520
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ1 Heifer

Sample and source description: WWQ1 Heifer

Sample date: 02/23/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	344.00	109.00	0.00	0.00	209.00	998.00	302.00	88.00	1,330.00	0.00	0.00	317.00	259.00	15,500.00	15,100
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ2 Heifer

Sample and source description: WWQ2 Heifer

Sample date: 04/12/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	272.00	86.80	0.00	0.00	256.00	1,480.00								13,300.00	13,400
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ2 Main

Sample and source description: WWQ2 Main

Sample date: 04/12/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	750.00	459.00	0.00	0.00	76.30	921.00								8,980.00	6,660
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Slurry

Sample and source description: Slurry

Sample date: 05/22/2023 Material type: Process wastewater sludge Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2,350.00	588.00	0.00	0.00	232.00	27.90								8,960.00	6,620
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ3 Heifer

Sample and source description: WWQ3 Heifer

Sample date: 07/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	742.00	392.00	0.00	0.00	485.00	2,550.00								0.00	19,700
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ3 Main

Sample and source description: WWQ3 Main

Sample date: 07/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	736.00	504.00	0.00	0.00	33.80	1,010.00								9,930.00	8,980
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Slurry

Sample and source description: Slurry

Sample date: 10/20/2023 Material type: Process wastewater sludge Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	532.00	140.00	0.00	0.00	179.00	499.00								5,760.00	3,060
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ4 Dairy

Sample and source description: WWQ4 Dairy

Sample date: 11/01/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	616.00	428.00	0.00	0.00	44.50	593.00								7,780.00	5,180
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ4 Heifer

Sample and source description: WWQ4 Heifer

Sample date: 11/01/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,080.00	302.00	0.00	0.00	471.00	2,110.00								0.00	20,000
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

MID Lateral 5

MID Lateral 5

Sample description: MID Lateral 5

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								38.10	37
DL	1.00	0.50	0.50								10.00	10

OID

OID

OID

Sample description: OID

Sample date: 08/11/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	0
DL	0.50	0.50	0.50								1.00	10

Canal OID

Sample description: Canal OID

Sample date: 08/18/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.01	0.00	0.00								70.90	52
DL	1.00	0.50	0.50								10.00	10

D. SOIL ANALYSES

Field 1

Soil

Sample and source description: Soil

Sample date: 04/12/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			100.00				
DL			4.00				

Field 2

Field 2

Soil

Sample and source description: Soil

Sample date: 04/12/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			52.00				
DL			4.00				

Field 3

Soil

Sample and source description: Soil

Sample date: 04/12/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			68.00				
DL			4.00				

Field 4

Soil

Sample and source description: Soil

Sample date: 06/12/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			73.00				
DL			4.00				

Field 5

Field 5

Soil

Sample and source description: Soil

Sample date: 10/11/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			73.00				
DL			4.00				

Field 6

Soil

Sample and source description: Soil

Sample date: 09/22/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			86.00				
DL			4.00				

Field B1

Soil

Sample and source description: Soil

Sample date: 11/22/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			38.00				
DL			4.00				

Field B2

Field B2

Soil

Sample and source description: Soil

Sample date: 06/12/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			31.00				
DL			4.00				

Field B3

Soil

Sample and source description: Soil

Sample date: 11/22/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			84.00				
DL			4.00				

Field B4

Soil

Sample and source description: Soil

Sample date: 11/22/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			84.00				
DL			4.00				

Field B5

Field B5

Soil

Sample and source description: Soil

Sample date: 11/22/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			38.00				
DL			4.00				

Squirrel Field 1

Soil

Sample and source description: Soil

Sample date: 06/12/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			15.00				
DL			4.00				

Squirrel Field 2

Soil

Sample and source description: Soil

Sample date: 06/12/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			16.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

Field 1 - 04/14/2023: Corn, silage

Field 1 - 04/14/2023: Corn, silage

Corn SIlage

Sample and source description: Corn SIlage

Sample date: 08/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 53.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,990.00	870.00	6,010.00		6.60
DL	100.00	100.00	100.00		0.01

Field 1 - 08/20/2023: Sudangrass, silage

Sudan

Sample and source description: Sudan

Sample date: 11/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,500.00	900.00	6,000.00		0.00
DL	100.00	100.00	100.00		0.01

Field 2 - 04/14/2023: Corn, silage

Corn SIlage

Sample and source description: Corn SIlage

Sample date: 08/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,100.00	650.00	6,120.00		7.30
DL	100.00	100.00	100.00		0.01

Field 2 - 04/14/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,040.00	420.00	10,100.00		6.50
DL	100.00	100.00	100.00		0.01

Field 2 - 08/20/2023: Sudangrass, silage

Sudan

Sample and source description: Sudan

Sample date: 11/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,500.00	900.00	6,000.00		0.00
DL	100.00	100.00	100.00		0.01

Field 3 - 06/12/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 08/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,630.00	600.00	5,070.00		11.00
DL	100.00	100.00	100.00		0.01

Field 4 - 11/30/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 81.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,480.00	570.00	4,880.00		9.92
DL	100.00	100.00	100.00		0.01

Field 4 - 06/22/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,240.00	590.00	4,270.00		6.60
DL	100.00	100.00	100.00		0.01

Field 5 - 11/30/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 75.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,750.00	530.00	4,720.00		8.57
DL	100.00	100.00	100.00		0.01

Field 5 - 06/20/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/11/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,630.00	380.00	6,460.00		6.10
DL	100.00	100.00	100.00		0.01

Field 6 - 11/30/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,530.00	640.00	5,640.00		9.37
DL	100.00	100.00	100.00		0.01

Field 6 - 06/12/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,100.00	560.00	4,960.00		6.70
DL	100.00	100.00	100.00		0.01

Field B1 - 02/10/2013: Almond, in shell

Almonds

Sample and source description: Almonds

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 17.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,800.00	1,180.00	9,300.00		11.30
DL	100.00	100.00	100.00		0.01

Field B2 - 10/31/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 52.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,200.00	860.00	6,550.00		6.90
DL	100.00	100.00	100.00		0.01

Field B2 - 06/19/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,490.00	370.00	5,030.00		6.70
DL	100.00	100.00	100.00		0.01

Field B3 - 01/01/2023: Pasture

Pasture

Sample and source description: Pasture

Sample date: 09/07/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 8.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,660.00	1,040.00	10,400.00		4.80
DL	100.00	100.00	100.00		0.01

Field B4 - 01/01/2023: Pasture

Pasture

Sample and source description: Pasture

Sample date: 09/07/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 9.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,090.00	1,170.00	15,800.00		7.91
DL	100.00	100.00	100.00		0.01

Field B5 - 01/01/2023: Pasture

Pasture

Sample and source description: Pasture

Sample date: 09/07/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 10.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,000.00	1,800.00	15,900.00		7.66
DL	100.00	100.00	100.00		0.01

Squirrel Field 1 - 02/05/2010: Almond, in shell

Almonds

Sample and source description: Almonds

Sample date: 10/11/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 31.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,300.00	1,540.00	13,300.00		11.30
DL	100.00	100.00	100.00		0.01

Squirrel Field 2 - 11/30/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,730.00	560.00	3,920.00		8.50
DL	100.00	100.00	100.00		0.01

Squirrel Field 2 - 06/16/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,930.00	480.00	2,860.00		6.00
DL	100.00	100.00	100.00		0.01

Squirrel Field 2 - 06/16/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,680.00	880.00	7,330.00		6.90
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

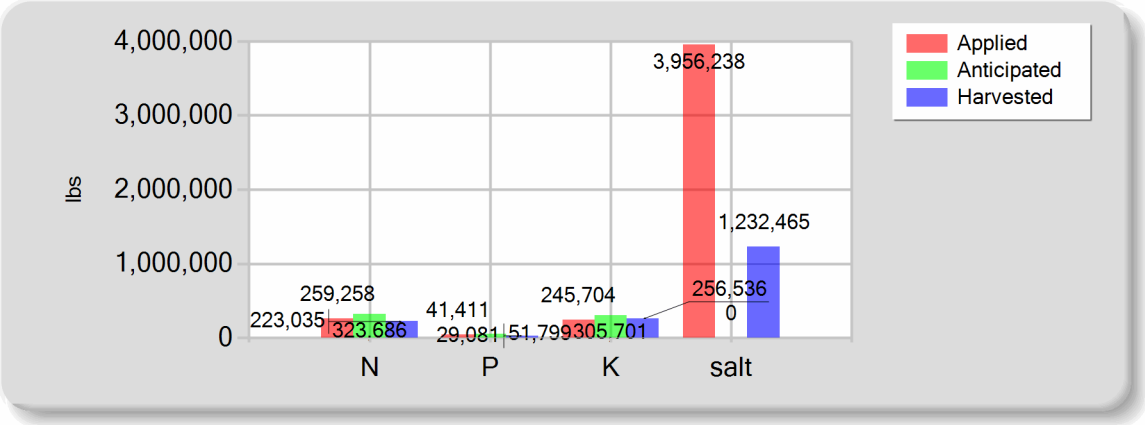
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

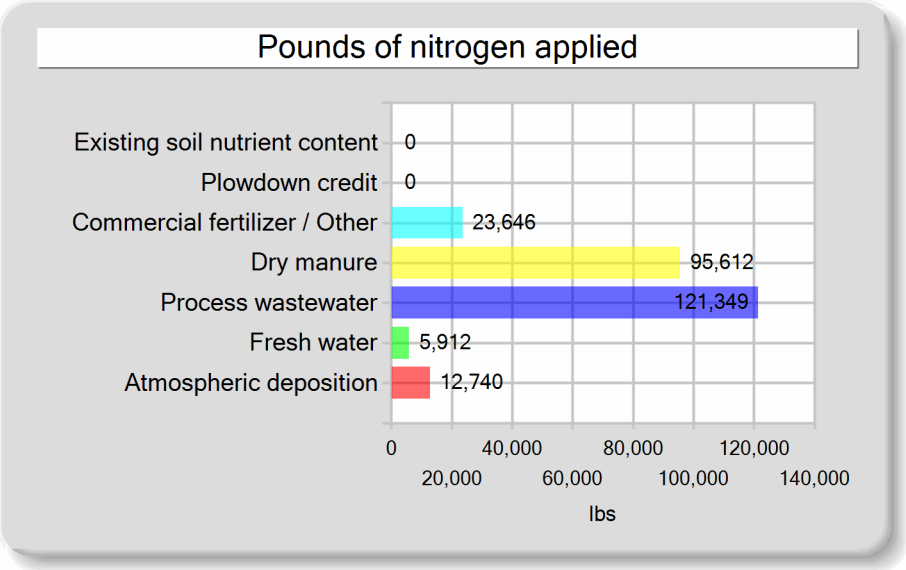
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

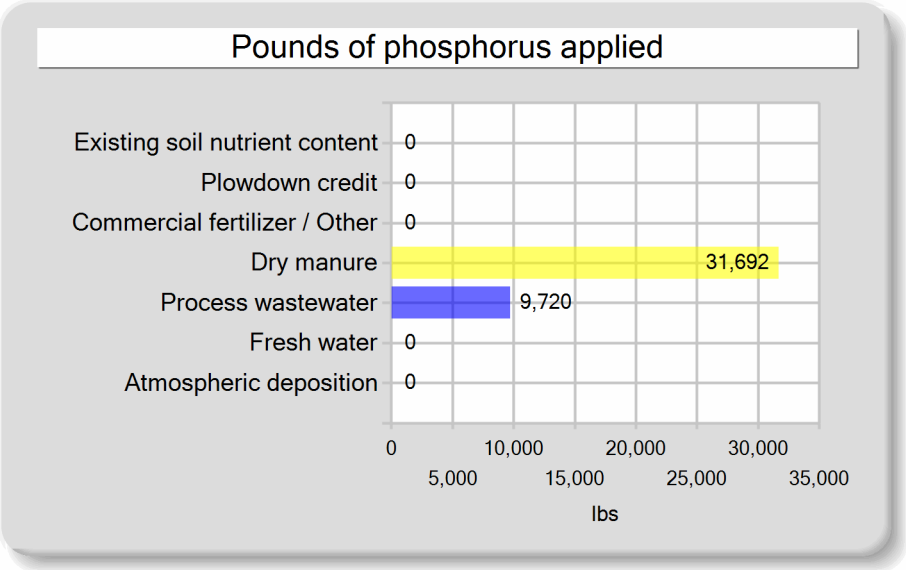
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	23,645.60	0.00	0.00	0.00
Dry manure	95,611.76	31,691.62	135,870.68	2,644,644.12
Process wastewater	121,348.95	9,719.55	109,833.42	928,877.74
Fresh water	5,911.85	0.00	0.00	382,715.96
Atmospheric deposition	12,740.00	0.00	0.00	0.00
Total nutrients applied	259,258.15	41,411.17	245,704.10	3,956,237.82
Anticipated crop nutrient removal	323,686.00	51,799.00	305,701.00	0.00
Actual crop nutrient removal	223,034.94	29,081.44	256,535.86	1,232,465.03
Nutrient balance	36,223.21	12,329.73	-10,831.76	2,723,772.79
Applied to removed ratio	1.16	1.42	0.96	3.21

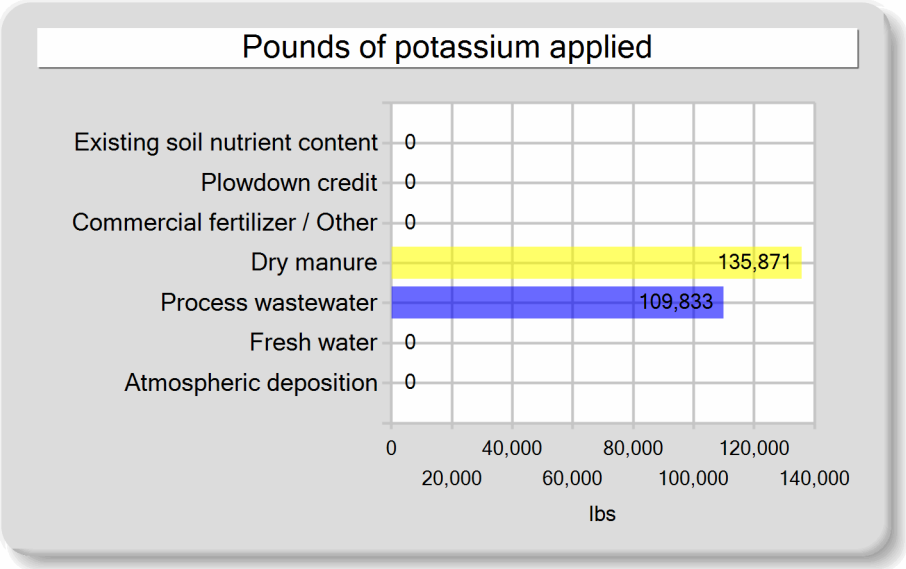
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

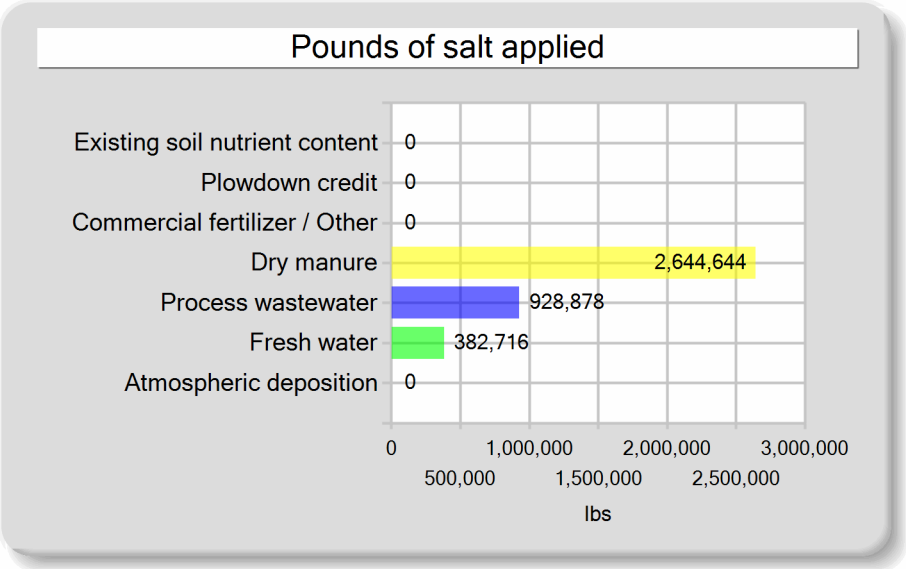


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Field 5 oat silage (1.47) and Field Squirrel 2 oat silage (1.84) had high nitrogen field ratios as crops failed to meet anticipated nitrogen removal of 160 pounds/acre. The subject fields removed 101.1 and 95.5 pounds of nitrogen/acre respectively. Yields for the fields were adequate however a low nitrogen value from the plant tissue lab analysis resulted in only 5.5 and 5.46 pounds of nitrogen removed per harvested ton compared to the anticipated 10 pounds of nitrogen removed per harvested ton which was the ultimate reason for the high nitrogen field ratios. Nitrogen application to the fields was not considered excessive but in-line with that recommended in the dairy's nutrient management plan.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Willem Postma
Willem Postma (Jun 27, 2024 12:33 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Willem Postma	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 27, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

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40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Willem Postma Dairy #1
P.O. Box 576529
Modesto, CA 95357

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/22/23 1:30 PM
SUBMITTED BY: Jeanette
ANALYZED DATE: 8/25/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 18.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 1	T3234028A	Electrical Conductivity (EC)	EPA 120.1 and	0.655	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	14.8	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	419	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.0	95.2 (T3234028A)	95.2 (T3234028A)	0.000
Electrical Conductivity (EC)	98.4			2.53
Nitrate Nitrogen (NO3-N)	103	99.9 (W3235002A)	99.9 (W3235002A)	0.060

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

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Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/22/23 1:30 PM
SUBMITTED BY: Jeanette
ANALYZED DATE: 8/25/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 18.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 2	T3234028B	Electrical Conductivity (EC)	EPA 120.1 and	0.656	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	15.0	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	420	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.0	95.2 (T3234028A)	95.2 (T3234028A)	0.000
Electrical Conductivity (EC)	98.4			2.53
Nitrate Nitrogen (NO3-N)	103	99.9 (W3235002A)	99.9 (W3235002A)	0.060

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P.O. Box 576529
Modesto, CA 95357

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/22/23 1:30 PM
SUBMITTED BY: Jeanette
ANALYZED DATE: 8/25/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 18.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 3	T3234028C	Electrical Conductivity (EC)	EPA 120.1 and	0.257	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	3.02	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	164	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.0	95.2 (T3234028A)	95.2 (T3234028A)	0.000
Electrical Conductivity (EC)	98.4			2.53
Nitrate Nitrogen (NO3-N)	103	99.9 (W3235002A)	99.9 (W3235002A)	0.060

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DENELE ANALYTICAL, INC.
865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Willem Postma Dairy
Mailing address: P.O. Box 576529
(not facility address) Modesto CA 95357
Phone#: 209-551-2930

Authorized Copy Released To: F&R Ag.
Sampled By: D. Bygrave
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

*Required Field Measurement
___ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	8-22-23	11:00am	DW1	domestic well	18.0°		✓				
2	8-22-23	11:00am	DW2	"	18.8°		✓				
3	8-22-23	11:15am	DW3	"	18.0°		✓				
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time
1st	Jennifer Sawyer	F&R		
2nd		DENELE ANALYTICAL, INC.	8-22-23 1:30	

(LABORATORY USE ONLY)
Logged In By:
LABORATORY NUMBER: T3234028

Required to be submitted to any agency? (YES (NO)

Willem Postma Dairy #1 2023 Annual Report

Final Audit Report 2024-06-27

Created:	2024-06-27
By:	Vince Furtado (vfurtado@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAeT1s9jgInDYEiaVepvUpA3YtJSgZ6pqF

"Willem Postma Dairy #1 2023 Annual Report" History

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2024-06-27 - 5:40:21 PM GMT

Document emailed to wpostma@att.net for signature
2024-06-27 - 5:41:21 PM GMT

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